Curriculum Vitae

Name : Apisit Tongchai

Born: April 5, 1980, Yasothon Province, Thailand

E-mail: a.tongchai@gmail.com, atong@ipst.ac.th

Website: https://apisit-man.github.io/

Current address:

1959/324 Dcondo Sukhumvit 109, SamrongNue, Samutprakarn, Thailand, 10270

Education:

- Ph.D. (Science and Technology Education) Mahidol University, Thailand (Scholarship from IPST, Ministry of Education) (200 - 2009)
- Diploma degree (Teaching Profession)
  KhonKean Universtiy, Thailand (Scholarship from IPST, Ministry of Education) (2003)
- B.Sc. Physics (2nd Hons.)
  Ubonratchathani University, Thailand (Scholarship from IPST, Ministry of Education) (1999 - 2002)

Ph.D. thesis topic (Adviser: Asst. Prof. Dr. Kwan Arrayathnitkul)

- Development and Application of Mechanical Wave Conceptual Test to Diagnose Students' conceptions

Overseas Research Experience

- Learning and Development Program | Dublin, Ireland (Mar–Apr 2025) Awarded the OCSC Government Personnel Development Scholarship to participate in an intensive leadership and organizational growth program.
- STEM Education Study Visit | Taiwan (Apr 2023). Observed advanced pedagogical practices and integrated STEM curricula, organized by the Science, Mathematics, and Technology Teachers Association of Thailand (SMT).
- Professional Development for In-Service Teachers | Jerusalem, Israel (May 2017). Selected for a specialized pedagogical workshop by the Embassy of Israel, focusing on innovative teaching methodologies.

- Sakura Science Plan: STEM Education Training | Saitama University, Japan (Feb 2016). Completed advanced STEM training and research exchange under a scholarship from the Japan Science and Technology Agency (JST).
- STEM & Teacher Development Study Visit | Vietnam (Jan 2015). Engaged in cross-cultural knowledge exchange regarding science education standards, organized by the Science and Technology Teachers Association of Thailand.
- 76th ITEEA Annual Conference | Orlando, Florida, USA (2014). Represented Thailand at the International Technology and Engineering Educators Association global forum for STEM education.
- NSTA's STEM Forum and Expo | Missouri, USA (2013). Participated in academic sessions and study visits focused on global science education trends through the National Science Teaching Association.
- 73rd ITEEA Annual Conference | Minnesota, USA (2011). Served as an international delegate for technology and engineering educator seminars and networking.
- 2nd ASEAN+3 Workshop & Student Camp for the Gifted | South Korea (July 2010). Completed a regional program at Kyungnam University dedicated to nurturing excellence in gifted science education.

International Conference Participations

- Participated in the 76th International Technology and Engineering Educator Association (the 76th ITEEA) during 26 – 29 March 2014 held in Orlando, Florida, USA
- Participated in the NSTA's STEM Forum and Expo 2013 held in St. Louis Missouri, USA, May 2013.
- Participated in the 73rd International Technology and Engineering Educator Association (the 73rd ITEEA) during 24 - 27 March 2011, Minneapolis, Minnesota, USA.

Work experience:

October 2023 – Present

Academic specialist, Division of Technology, The Institute for the Promotion of Teaching Science and Technology (IPST), Ministry of Education. Job description: Research and Development on the Design and Technology subject including STEM education as well as Artificial Intelligence in K-12 Education Develop student learning books, teaching guides, and do teacher training.

May 2018 – September 2023

Specialist, Division of Teacher and Educational Personnel Development. The Institute for the Promotion of Teaching Science and Technology (IPST), Ministry of Education. Job description: Research and development about Teacher Professional development in Science, Mathematics and Technology as well as STEM Education in Basic education.

June 2017 – April 2018

Academic staff, Technology Division, The Institute for the Promotion of Teaching Science and Technology (IPST), Ministry of Education. Job description: Develop and research about the design and technology national curriculum, develop learning books and teacher guides on the design and technology subject as well as doing STEM Education professional development.

February 2014 – May 2017

Academic staff, National STEM Education Center, The Institute for the Promotion of Teaching Science and Technology (IPST), Ministry of Education. Job description: Designing STEM learning activities, STEM Education standards, and doing STEM Education professional development.

September 2009 – January 2014

Academic staff, Department of Design and Technology Education, The Institute for the Promotion of Teaching Science and Technology (IPST), Ministry of Education. Job description: Developing textbooks in Design and Technology Education. Doing research on teaching and learning Design and Technology. Doing Professional development in Science and Technology.

Selected Publications

Tongchai, A. (2026). How to use AI as a "thought partner," not a "replacement thinker". IPST Magazine, 54(258). (in Thai).

Tongchai, A., Chuamklang, S., Rattanatippayaporn, B., Suksai, S., Metametha, K., & Pojananukij, N. (2026). Teachers' Familiarity, Readiness, and Anticipated Benefits and Barriers of Generative AI in Design and Technology Education. *Journal of Social and Scientific Education, 3*(1), 15–26. https://doi.org/10.58230/josse.v3i1.446

Tongchai, A. (2025). STEM teachers' readiness and behavioral intentions toward generative AI: Insights from a Thai professional development workshop. *Journal of Science and Mathematics Education in Southeast Asia, 48*, 88–108.

Tongchai, A., Malakul, S. (2025). Thai Teachers'Perceptions of Integrating Generative AI in K-12 Education: Opportunities and Challenges. *TechTrends*. https://doi.org/10.1007/s11528-025-01128-3

Tongchai, A. (2023). Single -Point Rubric: An Alternative Tool of Assessment for Learning. *The Institute for the Promotion of Teaching Science and Technology Magazine*. 51 (241), 39-44. (in Thai).

Tongchai, A. (2022). Enhancing and assessing students' higher order thinking. *Journal of The Science, Mathematics and Technology Association of Thailand*. 30 (Sep-December), 3-11. (in Thai).

Tongchai. A. (2022). Paper Rocket Challenge. *Technology and Engineering Teacher. 82(1).*

Apisit Tongchai, Patcharee R. Wichaidit. (2020). The Analysis of SMT Teachers' Self-Efficacy towards Enhancing Student Thinking and Problem-Solving Skills: Preliminary Study. *Journal of Science and Mathematics Education in Southeast Asia, 43.*

Apisit Tongchai, Patcharee R. Wichaidit, & Numphon Koocharoenpisal. (2019). A Professional Development Program to Enhance Thinking and Problem Solving Skills for Thai Science, Mathematics and Technology (SMT) Teachers. *Journal of Science and Mathematics Education in Southeast Asia,* 42(1), 1-25.

Tongchai, A. (2016). The importance of Engineering in Teaching Science in the 21st century. *Kasetsart Educational Review*. 31 (3), 48-53. (in Thai).

Tongchai, A., Sharma, M. D., Johnston, I. D., Arayathanitkul, K., & Soankwan, C. (2011). Consistency of students' conceptions of wave propagation: Findings from a conceptual survey in mechanical waves. Physical Review: Physics Education Research. *Phys. Rev. ST Phys. Educ. Res. 7*.

Tongchai, A., Sharma, M. D., Johnston, I. D., Arayathanitkul, K., & Soankwan, C. (2008). Developing, evaluating and demonstrating the use of a conceptual survey in Mechanical Waves. *International Journal of Science Education*.

Research expertise:

- Professional Development in Science and Technology Education in the 21st century learning
- Development of Integrated Science, Technology, Engineering and Mathematic (STEM) Education
- Artificial Intelligence in Education.